Predicting Rain Tomorrow in Australia

An explainable ML model for travel planning decisions

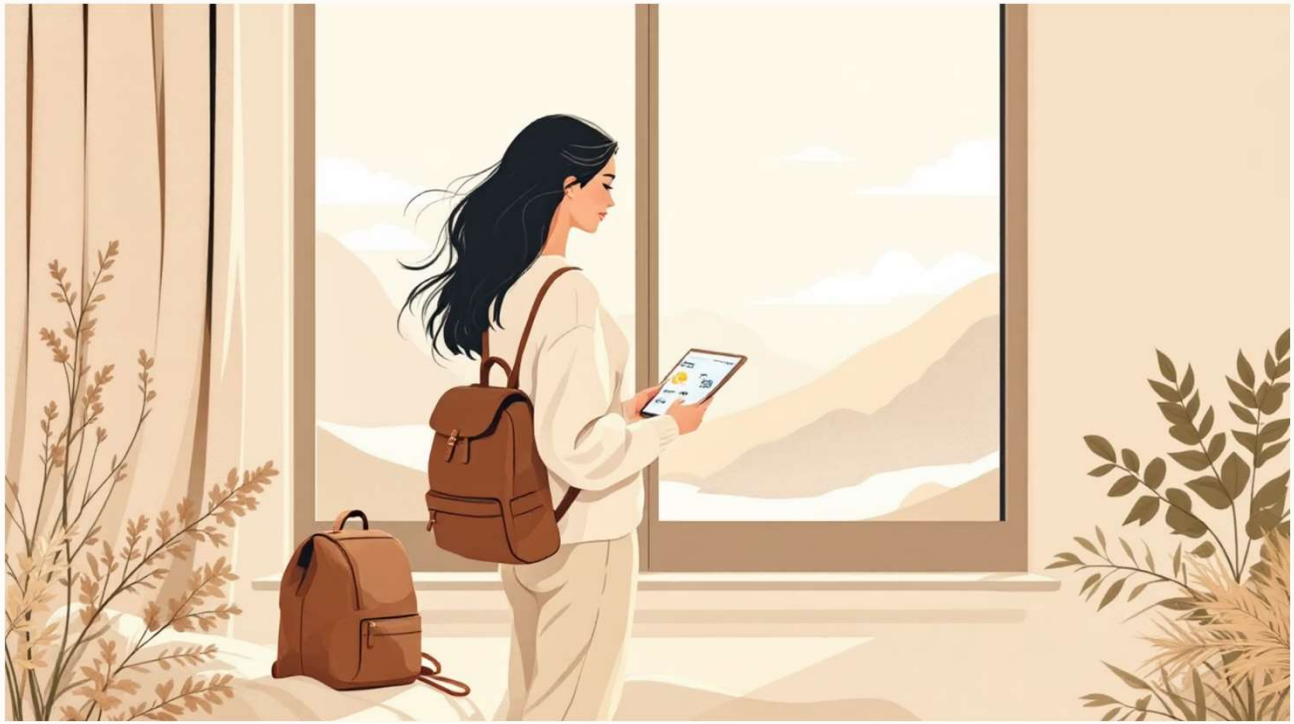
The Traveler's Question

The Core Question

"Will it rain tomorrow?" (Yes/No) — a binary prediction that shapes travel plans

Why Travelers Care

- Planning outdoor activities and routes
- Packing appropriate gear and clothing
- Avoiding costly cancellations
- Optimizing time and budget



Model Output

Probability of rain + alert threshold triggers

Dataset Snapshot

133k

Daily Observations

Rows of weather data

11

Features

Variables including target

78%

No Rain

Majority class

22%

Rain Tomorrow

Minority class

Source: "Rain in Australia" Kaggle dataset — daily weather observations across Australian weather stations

Feature Engineering Choices

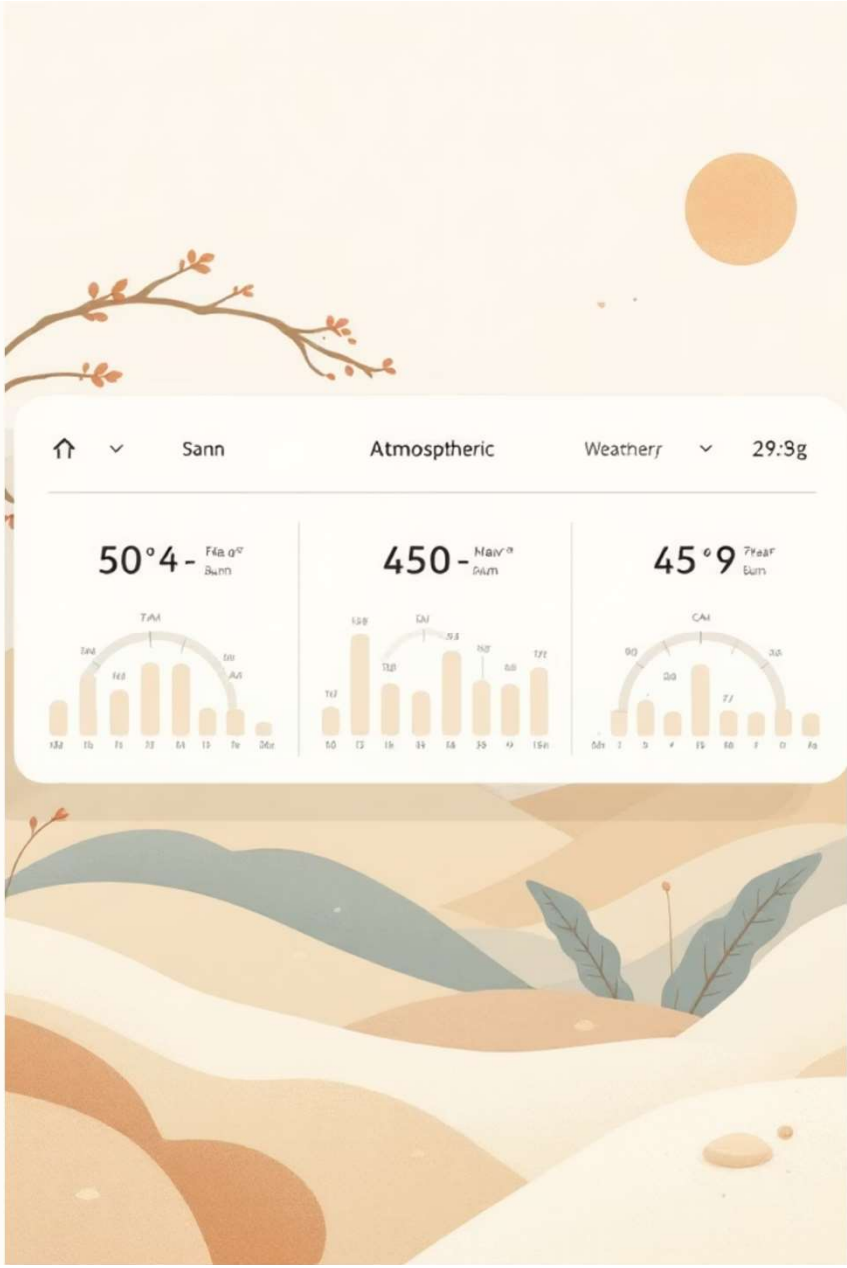
Features We Used

- Atmospheric signals: humidity, pressure, cloud cover
- Wind speed and direction
- Temperature readings (average)
- Daylight: sunshine and solar exposure
- Seasonality: month indicator

Features We Excluded

- **RainToday/Rainfall:** Persistence shortcuts excluded on purpose
- **Location:** Removed to train a general model

Trade-off: May reduce accuracy but improves generality across locations



Data Preparation Process

01

Missing Values

Pragmatic handling to keep dataset usable

02

Feature Engineering

Combine 9am/3pm readings into daily averages + month seasonality

03

Train/Test Split

Random stratified (75/25) to preserve class balance

04

SMOTE Application

Synthetic oversampling applied on training set only (not test)

How We Measure Success

Why Accuracy Misleads

With 78% No Rain class, a model that always predicts "No Rain" achieves 78% accuracy — useless for travel planning

Primary Metrics

- **Balanced Accuracy:** Treats both classes fairly
- **Macro F1:** Balances precision and recall

Goal

Improve recall for "Rain=Yes" while controlling false alarms

Three Models Compared

+=x

Logistic Regression

Interpretable baseline

Random Forest

Non-linear baseline

Gradient Boosting

Strong tabular model

Best Model Performance

Model Summary

Winner model: Random Forest

Balanced Accuracy

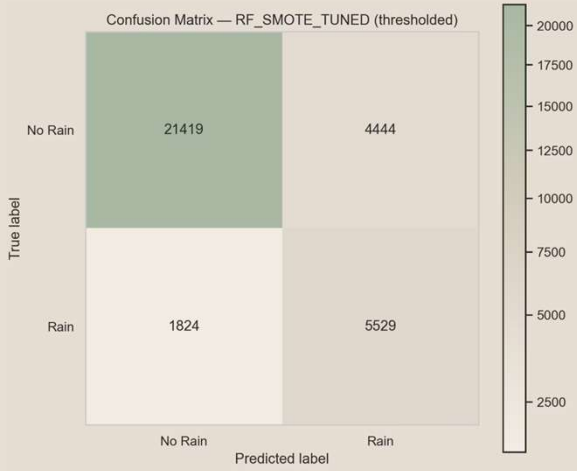
Macro F1

Macro Recall

Key Metric

Minority-class recall (Rain=Yes): 75%

Critical for travel alerts — catching rain days matters most



What Drives Rain Risk?

SHAP Explainability Analysis

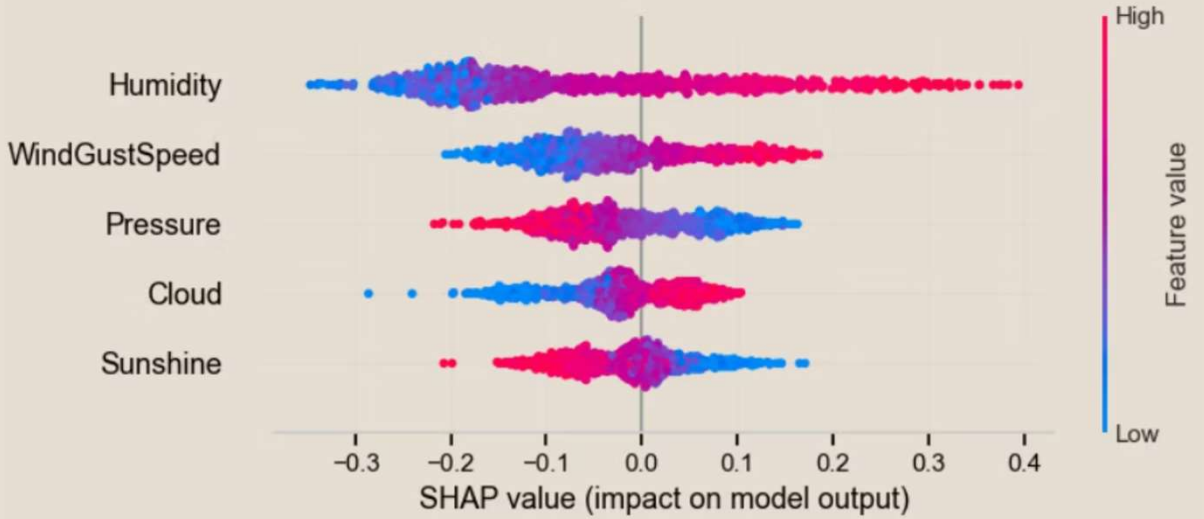
Feature importance from SHAP summary plot:

Higher Rain Probability

- Higher humidity
- More cloud cover
- Stronger wind gusts

Lower Rain Probability

- Higher pressure
- More sunshine
- Clearer skies



Limitations & Next Steps

Current Limitations

- Random split (not time-based)
- Location removed
- Pragmatic missing-value handling

Future Improvements

- Time-based validation
- Compare "with vs without persistence features"
- Calibrate probabilities

> An explainable rain-risk predictor that supports better travel planning decisions

Thank You!

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